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DESCRIPTION

These instructions explain how to take an A²B® project built in SigmaStudio® and download it to an A²B master board so that it self-boots with that program. For a board to self-boot when power is applied, the appropriate jumper or switch needs to be set/removed. On the legacy EVAL-AD24210WDZ A²B master board, the S1 switch must be in the disable position for the board to be controlled by SigmaStudio, and it must be in the enabled position to self-boot. On the EVAL-AD2425WD1BZ and EVAL-AD2428WD1BZ A²B master boards, the JP4/J4 jumper must be installed for SigmaStudio control, and it must be removed for the board to self-boot.

Before these instructions can be followed, the *A2B_Algo.dll* file must be copied into the SigmaStudio program folder. By default, this folder should be *C:\Program Files\Analog Devices\SigmaStudio 4.2*. If SigmaStudio is open when the file is copied, it will need to be closed and re-launched to see the new *A2B toolbox* folder.

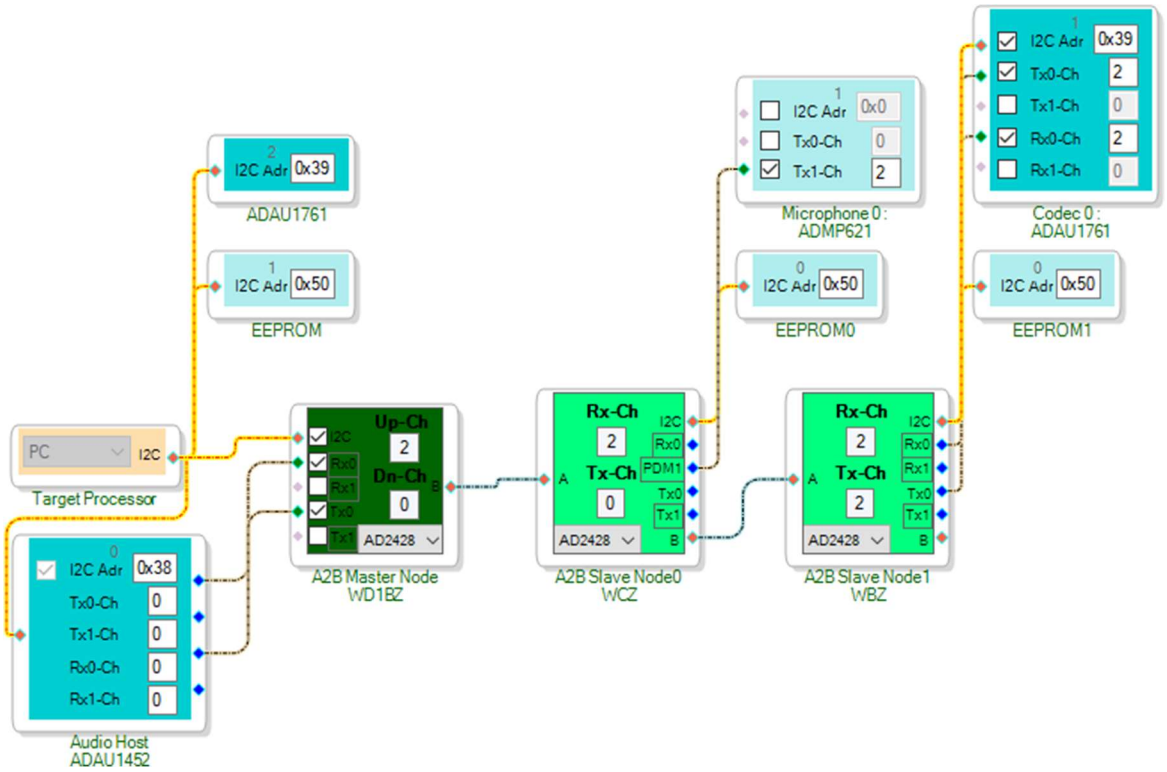


Figure 1. A²B Example Project

INSTRUCTIONS

1. From the A²B project file, do a *Link-Compile-Download* and verify that the project works as expected.



Figure 2. Link Compile Download Button

2. Right-click on the *Audio Host* block in the project and select *Peripheral Properties*. In the pop-up window, uncheck the option for 'Program during discovery' and then close the pop-up window.

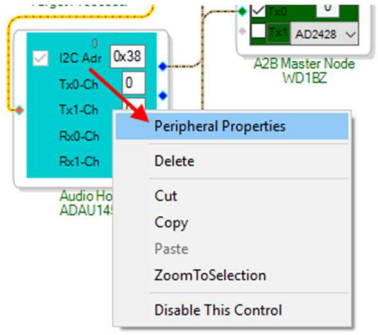


Figure 3. Host Context Menu, Peripheral Properties

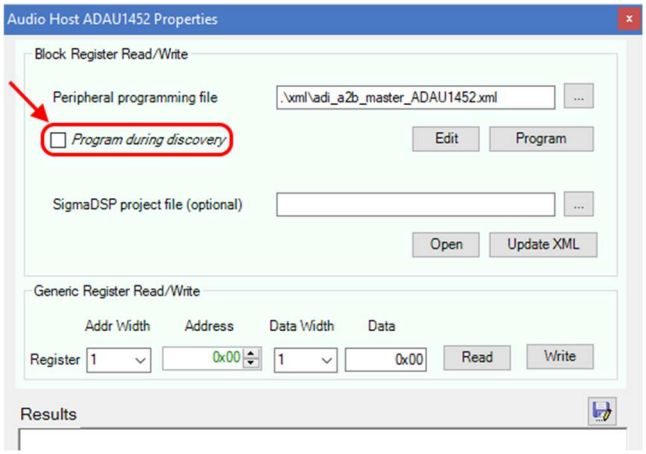


Figure 4. Host Peripheral Properties, Uncheck Program During Discovery

3. Do a second *Link-Compile-Download*. This will create an I²C command list (minus programming the host), which will be exported in the next step. If the host processor's SigmaStudio project performs I²C reads/writes to the A²B bus, then this *Link-Compile-Download* action may return errors (which can be ignored).

4. Right-click on the *Target Processor*, select *Export System Config Files*, and click on the 'I2C command list' tab.

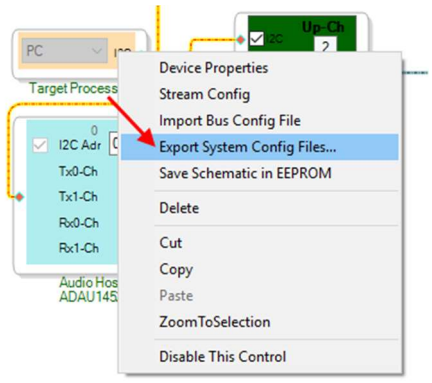


Figure 5. Target Processor Context Menu, Export System Config Files

5. Next to the 'File path for I2C Command list .XML file' dialog box, click the *Browse* button and specify a file name and location for where the command list is to be saved.

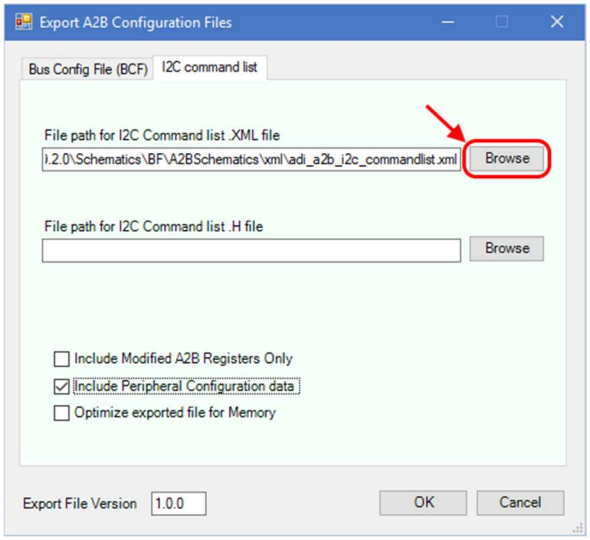


Figure 6. Export System Config Files, Browse File Location

6. Make sure the 'Include Peripheral Configuration data' checkbox is checked and click *OK*.

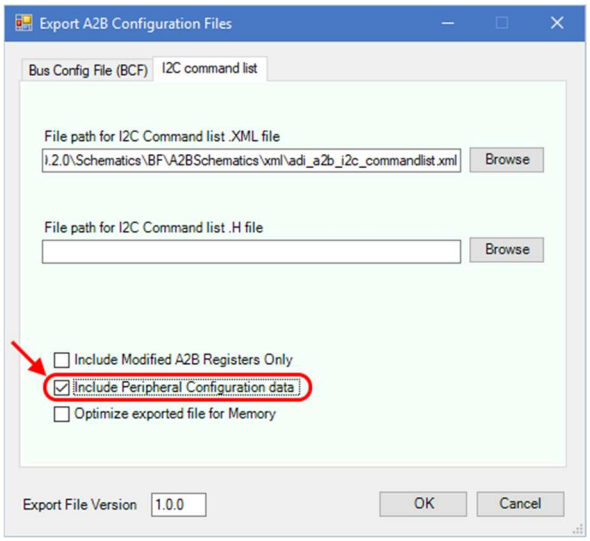


Figure 7. Export System Config Files, Include Peripheral Configuration Data

7. Open the audio host project file. If the file is linked to the *Audio Host* block, it can be accessed by right-clicking on the block and selecting *Peripheral Properties*. Below the 'SigmaDSP project file (optional)' dialog box, click the *Open* button.

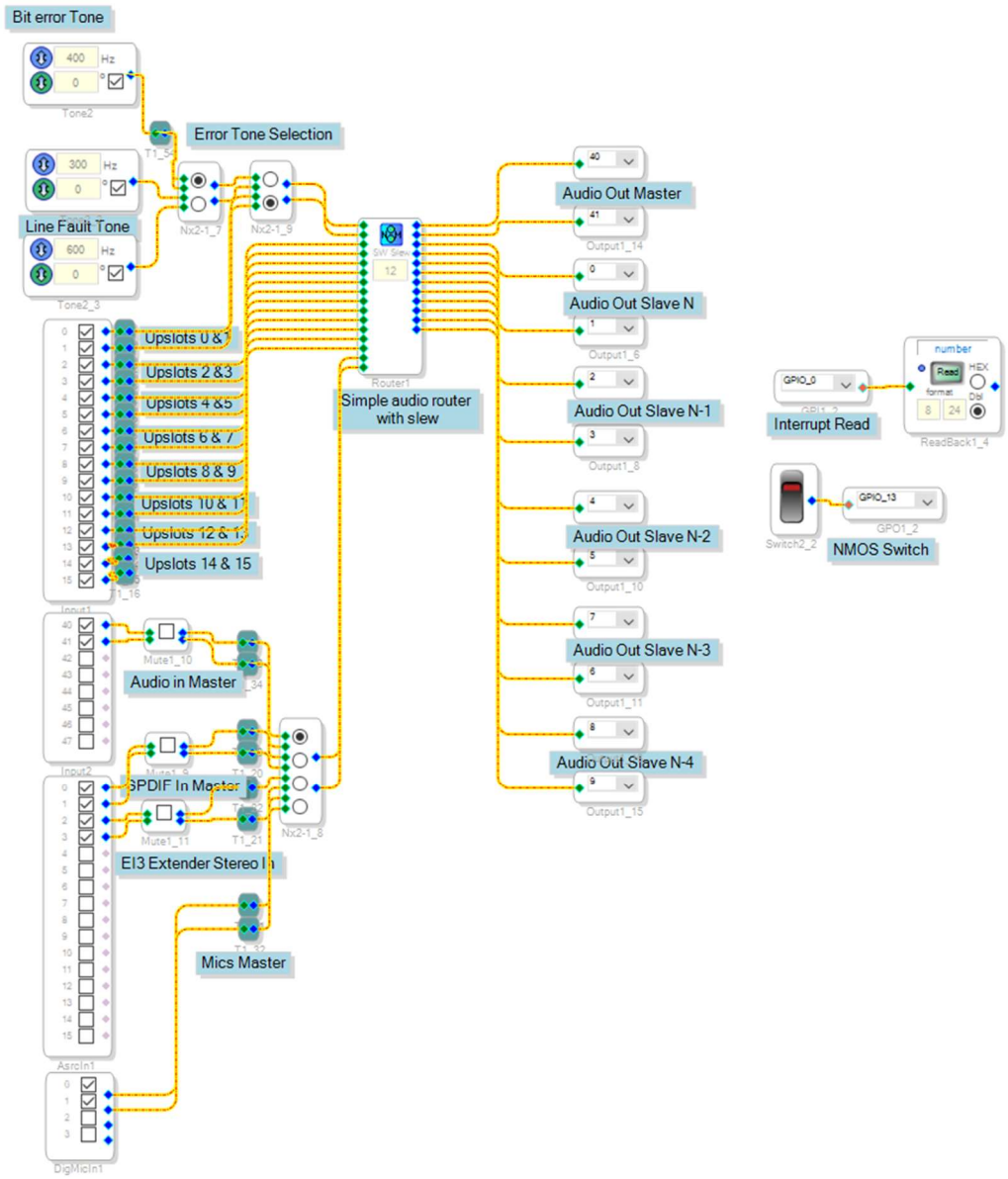


Figure 8. Example Host Project File

8. From the *Schematic* tab and the *Tree Toolbox*, drag and drop the *A2B > A2B > A2B Programmer* block onto the schematic.

   This step requires the A2B_Algo.dll to be installed in SigmaStudio. If not yet done, copy this file into the SigmaStudio program folder, which by default should be C:\Program Files\Analog Devices\SigmaStudio 4.2. SigmaStudio must be closed and re-launched to see the new toolbox items.

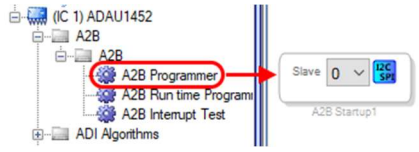


Figure 9. A²B Programmer Block Location

9. Click on the blue *I²C SPI* button on the *A²B Programmer* block to open the *Control Port Properties* window.

10. To the right of the *A²B Command List File* dialog box, click on the '…' button and select the file created in steps 5 and 6, then click *OK*.

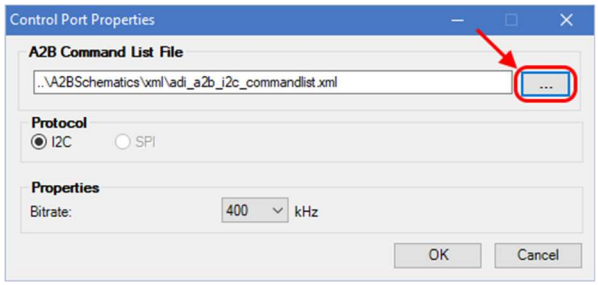


Figure 10. Control Port Properties Window, Select I2C Command List File

11. Do a *Link-Compile-Download* to verify that the *I²C Command List* file works. If it does not, go back to step 1 and try again.



Figure 11. Link Compile Download Button

12. Go to the *Hardware Configuration* tab, right-click on the *Audio Host IC* block, and select *Self boot Memory > Write Latest Compilation through DSP*.

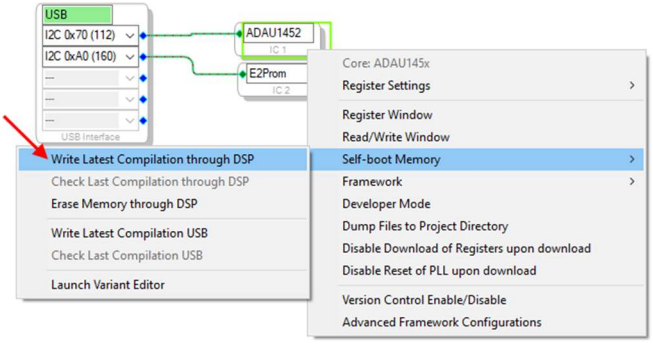


Figure 12. Write Latest Compilation Through DSP

13. On the window that pops up, just click *OK*. This step will take a few minutes to complete.

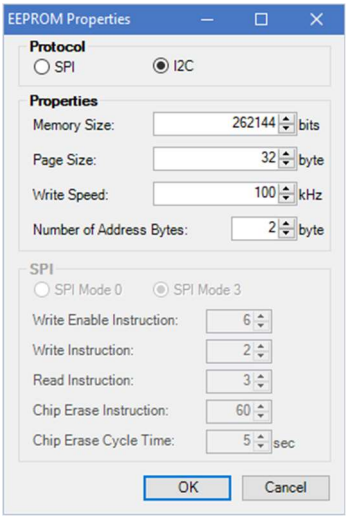


Figure 13. Write Compilation Settings

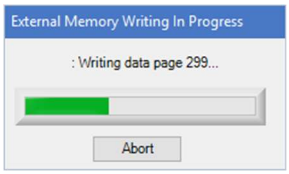


Figure 14. Download in Progress

14. Once the download has completed, remove the USBi programmer cable from the A²B master board and set the switch/jumper to enable self-boot mode.

15. Either cycle power to the A²B master board or press the RESET button. The A²B master board should now self-boot with the new A²B SigmaStudio program.

NOTES

I²C refers to a communications protocol originally developed by Philips Semiconductors (now NXP Semiconductors).



ESD Caution

ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

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